Optimized Double Band Spectra-Spatial Filters Coupled with Ternary ECOC for BCIs

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***Abstract*—Discriminative feature extraction for classification of motor imagery tasks to be deployed in EEG-based BCIs is still a bottleneck for the development of real-world applicable BCIs. Enhancing the performance of such systems requires optimization and enhancement in different signal processing steps. In this regard, this work demonstrates the effect of deploying optimized subject-specific spectral filters in the overall accuracy of the system. This effect is assessed while other processing units are kept on their most basic and common format. In other words, we propose a Bayesian optimization framework to optimize the spectral filters in a way that results in the most discriminant features and results in a high classification accuracy. On the other hand, we propose coupling the processing pipeline with the error correcting output coding (ECOC) classifiers. In the ECOC method, the classification problem is modeled as communication over a noisy channel where the misclassification error is corrected by error correcting techniques borrowed from information theory. In this work, we also propose a novel and modified version of the ECOC classifiers adopted to EEG classification problems which deploys ternary class codewords. The proposed framework is applied to a motor imagery classification problem and is evaluated over two datasets from BCI Competition which are named BCIC-IV_{2a} and BCIC-IV_{2b}, where the results indicate a noticeable enhancement over other methods developed for multi-class EEG classification.**

I. Introduction

Human brain, thanks to its vast and complex network of neurons, proposes the most powerful signal processing unit ever known and is capable of simultaneously analyzing and fusing multiple numbers of streaming signals from different modalities. Such capabilities of the brain have intrigued researchers to develop technologies which can directly communicate with the brain and consequently can provide further assistance/performance for the individuals. These technologies are called Brain-computer Interfaces (BCI) [1], [2] and are considered as the main building block of the human-in-the-loop, cyber-physical systems [3], [4]. In addition, BCIs are widely utilized in different engineering applications including assistive/rehabilitative systems [5]–[7], and neuro-prosthesis control [8]. The plasticity properties of the brain [9] have opened up various applications of BCIs in therapeutic fields and have significantly enhanced the effectiveness of rehabilitation therapies. Nowadays, rehabilitation-based BCIs [10] are of paramount importance in practical applications such as neurofeedback (NFB), therapy for autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD), schizophrenia, and motor rehabilitation for post-stroke patients to name a few. However, due to the complex behavior of the brain and various number of processes that are performed in real-time, the performance of the aforementioned systems rarely matches that of humans and the research in this field is still in its preliminary stages.

Every BCI system is composed of two building blocks which are typically called: (i) the signal recording or brain imaging module, and; (ii) the signal processing module. Based on the application of the BCI system, the former suggests different approaches to monitor the brain activity including the Electroencephalogram (EEG), Near Infra-red (NIR), Electrocorticogram (ECoG), or Magnetic Resonance Imaging (MRI). Having the affordability, portability and high temporal resolution taken into account as desired features of a brain imaging modality, EEG is the prime choice and is widely deployed in various BCI applications. The signal processing module aims at extracting meaningful information from the recorded signals and generally includes two major steps, i.e., "*feature generation,*" and "*feature translation.*" Feature generation performs the preprocessing and filtering on the signals and extracts the required features, and feature translation is the discussion on the features and assigns meaning to the features. The assigned meaning could be an action or a mental state or an emotion or anything else which the BCI is designed for. Next, we briefly review these two categories to better position the paper's contribution to the existing literature.

1. Feature Generation: When it comes to study the motory tasks based on the EEG signals, the sensorimotor activities [11] are believed to be the most informative representatives. The sensorimotor activities occur mainly in the frontal and parietal cortices of the brain and their spectral activity ranges from 8-30 Hz, where the contents in 8-13 Hz are considered as μ band and the ones in 13-30 Hz are named β band. In case of a voluntary movement, a slow drop in the power of the recorded signals from the motory cortices is observed which is best represented in the μ band, and once the action is done, a fast increase in the power of the signals is observed which is obvious in the β band. While the former event is called Event-related Desynchronization (ERD), the latter is named Event-related Synchronization (ERS). ERD and ERS are also known to be correlated with motor imagery (MI) tasks [12]. Among various processing techniques for sensorimotor activities, the Common Spatial Patterns (CSP) [13] is known to be the most powerful technique in detecting the ERD and ERS events. The CSP, which is basically developed for classification of two tasks, derives a linear transformation, known as a spatial filter, that extracts discriminative features

for the two tasks based on the ERD and ERS events. In other words, CSP aims at maximizing the variance of one MI task while minimizing the variance of the other task [14]. The superior power of the CSP approach in discriminating MI tasks has motivated the researchers [15]–[21] to extend the method further to enhance its classification performance. A number of these extensions include filter bank common spatial patterns (FBCSP) [16], regularized common spatial patterns (RCSP) [17], [18], separable common spatio-spectral patterns (SCSSP) [19], and optimized CSP based on fisher's Ratio [20].

Extensions to the CSP technique could be categorized into two major groups; the first group, including the works in [16] [19], is mainly based on the work in [16] which showed that decomposing the signals into several frequency bands using filterbanks significantly improves the performance of the CSP technique. The second group, including the work in [21], attempt to derive one optimized frequency band for each subject to boost the CSP performance.

2. Feature Translation: In case of dealing with classification of two classes of data, binary classifiers such as support vector machines (SVM) or linear discriminant analysis (LDA) are used. These methods are inherently designed for two-class problems and have shown superior performance in a different number of applications. On the other hand, when it comes to multi-class problems, these techniques are not directly applicable and new classifiers for multi-class problems should be deployed; either by devising new classifiers or modifying the current binary classifiers. The following two main approaches are used to address the multi-class problems: (i) Developing new classifiers, which are intrinsically designed for multi-class problems such as k-means, k-nearest neighbors (k-NN), and; (ii) Modifying the existing binary classifiers to fit multi-class problems.

Contribution: Although the filterbank methods have demonstrated promising results over different testing scenarios, a significant portion of the processing is wasted due to deploying fixed and general frequency bands. On the other hand and to the best of our knowledge, the methods utilizing the optimized spectral filters, have only inspected the optimization of a single frequency band. The paper addresses this gap. On the other hand and when it comes to feature translation, based on broad applications of binary classifiers such as the SVM or the LDA, it is observed that estimating a discriminant border for different classes is more productive than the estimation of the features' distribution (e.g., the k-means). This becomes more critical, especially when dealing with high dimensional features with limited data, which is the case in the EEG-based BCI systems. The paper also addresses this gap.

To address the foregoing drawback, we aim to couple filterbanks with optimized techniques to simultaneously take advantage of both solutions. In this regard, we propose a Bayesian framework to optimize the limits of two bandpass filters. To address the latter drawback, one can extend conventional binary classifiers (e.g., SVM and LDA) to multi-class settings (like multi-class SVM), or adapt current binary classifiers to become applicable to multi-class scenarios, e.g., popular techniques in this category are "one vs. one" and "one vs. all". In this paper, we propose an alternative solution, i.e.,

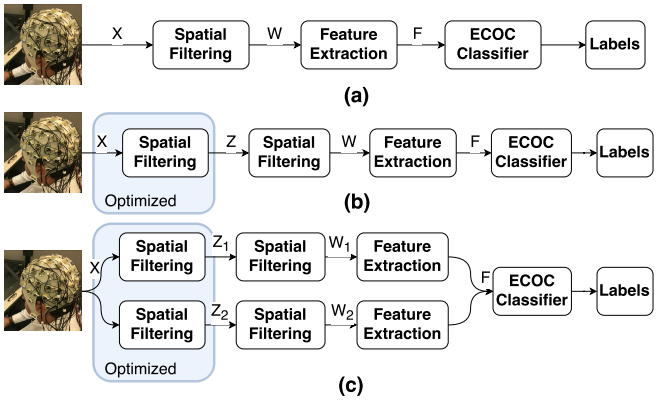


Fig. 1. The ECCSP framework: (a) The ECCSP. (b) The ECCSP-TB. (c) The ECCSP-TB2B.

to incorporate the error correcting output coding (ECOC) classifiers, which splits the classification problem into a number of binary ones. The ECOC [22] has been successfully used in other application domains, in particular, it has been effectively applied for text classification problems [23]. Despite the success of the ECOC in other fields, it has not yet been used for EEG classification in the BCI systems, which is one scope of this paper. In other words, we propose to deploy the ECOC and a modified version, we refer to as the *Ternary-ECOC*, as the classification component of the feature generation methods.

The paper proposes a novel processing framework for MI classifications tasks for BCI, referred to as the ECCSP. The overall theme of the proposed framework is demonstrated in Fig. 1 and is briefly outlined below:

1. ***The ECCSP***: A combination of ECOC with CSP is proposed, where we perform spatial filtering via CSP technique and introduce ECOC as the classifier (Fig. 1(a)).
2. ***The ECCSP-TB***: Improves each of the two underlying components of the ECCSP as follows: (i) We modify the ECOC algorithm and use its Ternary version, and; (ii) We incorporate a Bayesian framework to optimize the spatial-spectral filters prior to extracting the CSP features. The ECCSP-TB includes an optimization framework to obtain subject-specific spectral filters to enhance the classification accuracy, and updates the ECOC classifier to the proposed ternary version to further improve the classification accuracy (Fig. 1(b)).
3. ***The ECCSP-TB2B***: Further improves the ECCSP framework by using two optimized frequency bands for the spectral filtering instead of the one used in the ECCSP-TB version. The ECCSP-TB2B version improves the spectral filtering stage of the ECCSP-TB framework to extract subject and frequency specific features which lead to higher classification accuracy (Fig. 1(c)).

In summary, the paper makes the following key contributions: (i) Through the proposed ECCSP framework, we demonstrate the applicability of the ECOC classifiers for EEG studies and introduce a modified version of the ECOC which boosts the performance of the overall system; (ii) The proposed ECCSP framework integrates the idea of utilizing filter banks to extract more informative features with that of deriving one subject-specific and optimized spectral filter to analyze the MI signals, and; (iii) Finally, we illustrate the effect and importance of

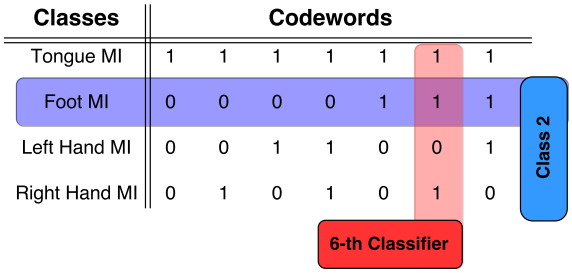


Fig. 2. The ECOC example with 7 bits for classifying 4 EEG classes.

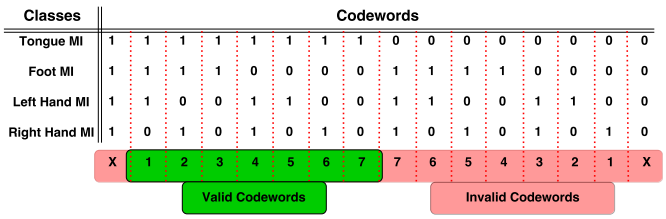


Fig. 3. All possible 4-bit binary codes.

utilization of the frequency specific features. By optimizing the spectral filter in the processing pipeline while keeping the feature extraction in its purest form, the performance is drastically improved. The rest of the paper is organized as follows: The ECCSP framework and its variants, ECCSP-TB and ECCSP-TB2B, are developed in Sections II, III and IV, respectively. Simulation results are provided in Section V. Finally, Section VI concludes the paper.

II. The ECCSP Framework

In this section, we present the main building blocks of the proposed ECCSP framework, which is developed by coupling the ECOC classification approach with the CSP feature extraction methodology. Later in Sections III and IV, we introduce different extended/modified variants of the ECCSP framework in terms of the ECCSP-TB and ECCSP-TB2B versions. For simplicity of the presentation, we use the following notations consistently: A scalar x is denoted by a non-bold letter; A vector $\boldsymbol{x}$ is represented by a lowercase bold letter, and; A matrix $\boldsymbol{X}$ is denoted by a capital bold letter. Transpose and trace of a given matrix $\boldsymbol{X}$ are denoted by $\boldsymbol{X}^T$, and $\mathrm{Tr}(\boldsymbol{X})$respectively. Finally, term $\mathbb{R}$ represents the real domain.

In this paper, our focus is on supervised learning (for classifying MI tasks) from N_{Trial} number of available training EEG epochs (trials) denoted by $\boldsymbol{X}_i \in \mathbb{R}^{N_{\text{ch}} \times N_{\text{t}}}$, for $(1 \leq i \leq N_{\text{Trial}})$. Trial $\boldsymbol{X}_i$ consists of N_{ch} number of EEG channel recordings, where N_{t} number of time samples are available per each channel. In other words, the processing is performed based on a training dataset denoted by $\{(\boldsymbol{X}_i, \Omega_i)\}$, where Ω_i is the correct label of the ith epoch, e.g., "MI of Tongue" or "MI of Right Foot hand." A pre-processing step, including bandpass filtering (commonly in the range of $0.5 - 100$Hz) and notch filtering, is typically applied prior to performing the MI classification task. The goal of the pre-processing is to extract the desired frequency contents of the EEG signals and to remove power line interferences.

A. ECOC Classifiers

The general ECOC classification for a problem with N_{c} number of classes is performed by first constructing a *binary coding matrix* denoted by $\boldsymbol{\mathcal{C}}$, which consists of N_{c} binary sequences each with N_{Bits} bits. Row n, for $(1 \leq n \leq N_{\text{c}})$, of the coding matrix $\boldsymbol{\mathcal{C}}_n$ represents a unique binary-vector or simply a unique codeword associated with class n. In the case of EEG classification with N_{c} classes of MI, N_{c} codewords of length N_{Bits} are used, where each codeword is assigned with a total number of N_{Bits} bits. Consequently, N_{Bits} classifiers ($\boldsymbol{\gamma}^{(j)}$, for $(1 \leq j \leq N_{\text{Bits}})$) are required. Adopted to the MI classification task based on EEG epochs, we use $3 \leq N_{\text{c}} \leq 7$ different classes, and construct codewords of length $N_{\text{Bits}} = 2^{N_{\text{c}}-1} - 1$. The codewords are built based on the following steps:

(i) The first row of the coding matrix consists of all zero entries.

(ii) Row $2 \leq i \leq N_{\text{c}}$ is constructed by concatenation of alternating runs of $2^{N_{\text{c}}-i}$ zero and one sequences.

In this setting, the binary classifier $\boldsymbol{\gamma}^j$ associated with the jth column of the coding matrix $\boldsymbol{\mathcal{C}}$ aims at distinguishing the trials which are labeled with 0 from the ones labeled with 1. This results in two super-sets for each classifier $\boldsymbol{\gamma}^j$ denoted by $\boldsymbol{S}^{(j,0)}$ and $\boldsymbol{S}^{(j,1)}$, which their corresponding features are extracted to be fed into the classifier. Training an ECOC classifier, therefore, comprises of learning a set $\boldsymbol{\Lambda} = \{\boldsymbol{\gamma}^{(1)}, \ldots, \boldsymbol{\gamma}^{(N_{\text{Bits}})}\}$ of independent binary classifiers.

Hypothesizing an unlabeled EEG epoch $\boldsymbol{X}_i$ is performed based on the set $\boldsymbol{\Lambda}$ of learned classifiers by first evaluating each classifier based on $\boldsymbol{X}_i$, i.e., to compute the binary decision of each classifier resulting in the generation of the following binary-vector

$$\boldsymbol{\Lambda}(\boldsymbol{X}_i) = \{\boldsymbol{\gamma}^{(1)}(\boldsymbol{X}_i), \ldots, \boldsymbol{\gamma}^{(N_{\text{Bits}})}(\boldsymbol{X}_i)\}. \tag{1}$$

Epoch $\boldsymbol{X}_i$ is assigned to a given class if the resulting bit-vector computed from Eq. (1) is similar to the codeword representing that class. In the likely case that the resulting bit-vector is not the same as any of the rows of the codebook $\boldsymbol{\mathcal{C}}$, trial $\boldsymbol{X}_i$ will be assigned to the closest codeword (in terms of the Hamming distance Δ) as follows

$$\boldsymbol{\Phi}(\boldsymbol{X}_i) = \mathrm{argmin}_m \Delta\big(\boldsymbol{\mathcal{C}}_m, \boldsymbol{\Lambda}(\boldsymbol{X}_i)\big), \tag{2}$$

where $\Delta(\boldsymbol{a}, \boldsymbol{b})$ is the number of bits in which vectors a and b differ. More specifically, first the Hamming distance between all the available codewords $\boldsymbol{\mathcal{C}}_i$ and the computed bit-vector $\boldsymbol{\Lambda}(\boldsymbol{X}_i)$ given by Eq. (1) is computed. Then epoch $\boldsymbol{X}_i$ is assigned to class n which has the minimum distance among the other codewords. Before presenting the ECCSP, below we present an example to better clarify the ECOC technique.

Illustrative Example: An illustrative example is shown in Fig. 2 where the goal is to classify given EEG epochs in to $N_{\text{c}} = 4$ categories, i.e., {Tongue MI, Foot MI, Left hand MI, Right hand MI}. The ECOC shown in this example consists of 7 classifiers ($N_{\text{Bits}} = 7$), the 6th of which is identified in Fig. 2 with a red rectangle. This 6th classifier tries to assign a given EEG epoch into either of the two super-sets, i.e., $\boldsymbol{S}^{(1)} = \{$Tongue MI, Foot MI, Right hand MI$\}$ and $\boldsymbol{S}^{(0)} = \{$Left hand MI$\}$. In other words, this classier will be

Algorithm 1 ECOC BASED EEG CLASSIFICATION

Input: $\{(\boldsymbol{X}_i, \Omega_i)\}_{i=1}^{N_{\text{Trial}}}$, and; N_{Bits}.
Output: N_{Bits} trained binary classifiers $\{\boldsymbol{\gamma}^{(1)}, \ldots, \boldsymbol{\gamma}^{(N_{\text{Bits}})}\}$.
S1. *Codebook Generation*: Form ($N_{\text{c}} \times N_{\text{Bits}}$) coding matrix $\boldsymbol{\mathcal{C}}$.
S2. *Classifier Design*:
for $j \in \{1, \ldots, N_{\text{Bits}}\}$ **do**
  Super-sets Construction: $\boldsymbol{S}^{(j,0)} = \{\boldsymbol{X}_i;\ \boldsymbol{\mathcal{C}}_{i,j} = 0\}$ and $\boldsymbol{S}^{(j,1)} = \{\boldsymbol{X}_i;\ \boldsymbol{\mathcal{C}}_{i,j} = 1\}$.
  Binary Classifier Construction: Train classifier $\boldsymbol{\gamma}^{(j)}$ discriminate between $\boldsymbol{S}^{(j,0)}$ and $\boldsymbol{S}^{(j,1)}$.
end for

trained to distinguish between trials labeled as "Tongue MI", "Foot MI", or "Right hand MI" and the trials labeled as "Left hand MI". In terms of our running example, we intuitively elaborate on the idea behind utilization of code words of length $2^{N_c-1}-1$ in the ECOC algorithm. Consider this scenario when $N_c = 4$ classes, first we try to find all the possible ways that labels 0 and 1 can be assigned to any of the classes. In practice, we are defining all the binary numbers that could be formed with 4 bits. Hence, we have 16 binary numbers between 0000 and 1111. Fig. 3 depicts all the 16 possible codewords. As it is labeled in Fig. 3, we are interested in the cases that both $\boldsymbol{S}^{(0)}$ and $\boldsymbol{S}^{(1)}$ super-sets could be formed, therefore, the first (all zero) and the last (all 1) cases are not applicable for the classification problem. Among the rest of the cases, it is observed that each case has its complement among the others. Algorithm 1 lists the required steps for classification of EEG epochs into different sets of MI tasks by using the ECOC classifiers. Next, we develop the feature extraction approach.

B. The ECOC-based CSP Spatial Filtering

The proposed ECCSP framework couples ECOC classification approach presented in Section II.B with CSP approach [15]. The latter forms a linear transformation matrix for projecting the EEG epochs into a lower-dimensional spatial subspace, which is then used by the former to construct N_{Bits} classifiers to learn the designed codebook $\boldsymbol{\mathcal{C}}$. Essentially, the CSP approach simultaneously diagonalize the normalized spatial covariance matrix of EEG epochs associated to the two super-sets to forms projection matrix that maximizes the variance of the two super-sets used by each of the N_{Bits} ECOC classifiers. Intuitively speaking, the reason that the CSP approach outperforms some well regarded analytical techniques (such as independent component analysis (ICA) or the principal component analysis (PCA)) for dimension reduction and feature extraction can be attributed to the fact that the CSP technique uses the labels of the data and handles the problem in a supervised fashion. Features required for classification/training of N_{Bits} binary classifiers are obtained by analyzing the spatially filtered signals.

Each classifier $\boldsymbol{\gamma}^{(j)}$, for ($1 \leq j \leq N_{\text{Bits}}$), within the ECCSP framework, uses its classifier-specific features. More specifically, for trial $\boldsymbol{X}_i$, at first we compute the spatially normalized covariance matrix as follow

$$\boldsymbol{\Sigma}_i = \frac{1}{N_{\text{t}} - 1}\big(\boldsymbol{X}_i - \boldsymbol{\mu}_i\big)\big(\boldsymbol{X}_i - \boldsymbol{\mu}_i\big)^T, \tag{3}$$

where $\boldsymbol{\mu}_i$ is the column-wise mean of $\boldsymbol{X}_i$. Typically, EEG signals are bandpassed resulting in the removal of the mean component, i.e., $\boldsymbol{\mu}_i = \boldsymbol{0}$. The full second-order statistical characteristics of the EEG epochs can then be inferred from

$$\boldsymbol{\Sigma}_i = \frac{\boldsymbol{X}_i \boldsymbol{X}_i^T}{\text{Tr}(\boldsymbol{X}_i \boldsymbol{X}_i^T)}. \tag{4}$$

For training classifier $\boldsymbol{\gamma}^{(j)}$, , for ($1 \leq j \leq N_{\text{Bits}}$), first we form its two corresponding super-sets $\boldsymbol{S}^{(j,0)}$ and $\boldsymbol{S}^{(j,1)}$, and then computer $\bar{\boldsymbol{\Sigma}}^{(j,0)}$ and $\bar{\boldsymbol{\Sigma}}^{(j,1)}$ by taking the average of all spatial covariance matrices computed based on Eq. (4) from all the epochs that belong super-sets ($\boldsymbol{S}^{(j,0)}$ and $\boldsymbol{S}^{(j,1)}$), respectively. To further clarify this step, let us describe it in terms of our illustrative example, here term $\bar{\boldsymbol{\Sigma}}^{(6,0)}$ is computed by taking the average of covariance matrices that belongs to the super-set $\boldsymbol{S}^{(0)} = \{\text{Left hand MI}\}$. Similarly, term $\bar{\boldsymbol{\Sigma}}^{(6,1)}$ is formed by taking the average of all normalized covariance matrices that belong to the second super-set which is $\boldsymbol{S}^{(1)} = \{$ Tongue MI, Foot MI, Right hand MI$\}$. The next step is to compute the composite spatial covariance matrix $\boldsymbol{\Sigma}^{(j,c)} = \bar{\boldsymbol{\Sigma}}^{(j,0)} + \bar{\boldsymbol{\Sigma}}^{(j,1)}$ and compute its eigenvalue decomposition as

$$\boldsymbol{\Sigma}^{(j,c)} = \boldsymbol{V}^{(j,c)} \boldsymbol{\lambda}^{(j,c)} [\boldsymbol{V}^{(j,c)}]^T, \tag{5}$$

where matrix $\boldsymbol{V}^{(j,c)}$ is constructed based on the eigenvectors of $\boldsymbol{\Sigma}^{(j,c)}$, and the diagonal matrix $\boldsymbol{\lambda}^{(j,c)}$ is constructed from eigenvalues of the composite matrix $\boldsymbol{\Sigma}^{(j,c)}$. These two terms are then used within the ECCSP framework uses to the following whitening transformation matrix

$$\boldsymbol{P}^{(j)} = \sqrt{\big[\boldsymbol{\lambda}^{(j,c)}\big]^{-1}} [\boldsymbol{V}^{(j,c)}]^T, \tag{6}$$

which equalizes the variance in the space spanned by the composite matrix's eigenvectors resulting in the eigenvalues of $\boldsymbol{P}^{(j)} \boldsymbol{\Sigma}^{(j,c)} [\boldsymbol{P}^{(j)}]^T$ to become one. The whitening matrix introduced in Eq. (6) is then used to transform matrices $\bar{\boldsymbol{\Sigma}}^{(j,0)}$ and $\bar{\boldsymbol{\Sigma}}^{(j,1)}$ (the average covariance matrices associated with each of the two super-sets) as

$$\boldsymbol{S}^{(j,1)} = \boldsymbol{P}^{(j)} \bar{\boldsymbol{\Sigma}}^{(j,1)} [\boldsymbol{P}^{(j)}]^T \text{and } \boldsymbol{S}^{(j,0)} = \boldsymbol{P}^{(j)} \bar{\boldsymbol{\Sigma}}^{(j,0)} [\boldsymbol{P}^{(j)}]^T. \tag{7}$$

The transformation introduced in Eq. (7) results in matrices $\boldsymbol{S}^{(j,0)}$ and $\boldsymbol{S}^{(j,1)}$ to have common eigenvectors given by $\boldsymbol{S}^{(j,1)} = \boldsymbol{B}^{(j)} \boldsymbol{\lambda}^{(j,1)} [\boldsymbol{B}^{(j)}]^T$ and $\boldsymbol{S}^{(j,0)} = \boldsymbol{B}^{(j)} \boldsymbol{\lambda}^{(j,0)} [\boldsymbol{B}^{(j)}]^T$, where the sum of the two eigenvalues equals an identity matrix $\boldsymbol{I}$ of appropriate dimension (i.e., $\boldsymbol{\lambda}^{(j,0)} + \boldsymbol{\lambda}^{(j,1)} = \boldsymbol{I}$). This neat property results in the eigenvector associated with the largest eigenvalue of $\bar{\boldsymbol{S}}^{(j,0)}$ to have the smallest eigenvalue associated with $\bar{\boldsymbol{S}}^{(j,1)}$ making the eigenvectors $\boldsymbol{B}^{(j)}$ appropriate useful for classifying the two super-sets. The whitened EEG epochs are then projected onto the eigenvectors associated with the largest and smallest eigenvalues of matrix $\boldsymbol{B}^{(j)}$ to form the optimal feature vectors (in the least square sense) to discriminate the two super-sets. The projection matrix associated with classifier $\boldsymbol{\gamma}^{(j)}$ is, therefore, computed as follows

$$\boldsymbol{W}^{(j)} = \big[\boldsymbol{P}^{(j)}\big]^T \boldsymbol{B}^{(j)}, \tag{8}$$

which is finally used to decompose each epoch $\boldsymbol{X}_i$ as

$$\boldsymbol{\mathcal{W}}_i^{(j)} = \big[\boldsymbol{W}^{(j)}\big]^T \boldsymbol{X}_i. \tag{9}$$

Algorithm 2 TRAINING IN THE ECCSP FRAMEWORK

Input: Training EEG Trials $\boldsymbol{X}$; The coding matrix: $\boldsymbol{\mathcal{C}}$.
Output: N_{Bits} trained binary classifiers $\{\boldsymbol{\gamma}^{(1)}, \ldots, \boldsymbol{\gamma}^{(N_{\text{Bits}})}\}$.

S1. According to the coding matrix $\boldsymbol{\mathcal{C}}$, construct the two super-sets for each classifier.
S2. Perform spatial filtering on the recordings by means of Eq. (9).
S3. Select the first and last m rows of the spatially filtered signals.
S4. Extract classifier-specific features using Eq. (10).

The jth classifier then constructs its feature vector $\boldsymbol{f}_i^{(j)}$ from the ith EEG trial $\boldsymbol{X}_i$ by only using the first and last m rows of $\boldsymbol{\mathcal{W}}_i^{(j)}$ [15] as follows

$$\boldsymbol{f}_i^{(j)} = \log\left(\frac{\text{var}(\boldsymbol{\mathcal{W}}_{i,p}^{(j)})}{\sum_{k=1}^{2m}\text{var}(\boldsymbol{\mathcal{W}}_{k,p}^{(j)})}\right), \tag{10}$$

where matrix $\boldsymbol{\mathcal{W}}_{i,p}^{(j)}$ contains the first and last m rows of matrix $\boldsymbol{\mathcal{W}}_i^{(j)}$ and var$(\cdot)$ denotes the variance operator. This completes the introduction to the proposed ECCSP framework summarized in Algorithm 2.

III. THE ECCSP-TB

In this section, we propose a modified variant of the ECCSP, referred to as the ECCSP-TB, which deploys a Bayesian framework to optimize the spectral filters within the ECCSP framework resulting in a set of class discriminative frequency bands and their associated spatial filters. In other words, the ECCSP-TB extends the spatial filtering of the ECCSP by utilization of an optimized spatial-spectral optimized filtering step and extends the multi-class classification approach by using a ternary version of the ECOC classifier. In the following sub-sections, we present details of the ECCSP-TB variant.

A. Bayesian Spectra-Spatial Filter Optimization

The ECCSP-TB performs spectral filtering and spatial filtering on the EEG signals as described below.

Spectral Filtering: This step is modeled as the convolution of the input EEG epoch $\boldsymbol{X}_i$ with the impulse response of a bandpass filter $h(k)$, i.e.,

$$\boldsymbol{\mathcal{Z}}_i = h \circledast \boldsymbol{X}_i, \tag{11}$$

where $\circledast$ denotes the convolution operation, and the cutoff frequency vector $\boldsymbol{\mathcal{B}}$ specifying the bandpass filter h is defined as

$$\boldsymbol{\mathcal{B}} \triangleq [b_s, b_e]^T. \tag{12}$$

In the ECCSP-TB framework, the cutoff frequencies are modeled as random variables and are optimized iteratively to increase the classification accuracy. In other words, we model the uncertainty in the cut-off frequencies of the spectral filters with a prior probability denoted by $p(\boldsymbol{\mathcal{B}})$ over random variable $\boldsymbol{\mathcal{B}}$. The prior density $p(\boldsymbol{\mathcal{B}})$ describes relative probabilities of different states (frequency bands) in which a single-trial EEG recording is correctly discriminated. The posterior probability distribution is then computed based on each bandpass-filtered single-trial EEG recording $\boldsymbol{\mathcal{Z}}_i$, for $(1 \leq i \leq N_{\text{Trial}})$, and its corresponding label denoted by Ω_i, as

$$p(\boldsymbol{\mathcal{B}}|\boldsymbol{\mathcal{Z}}_i, \Omega_i) = \frac{p(\boldsymbol{\mathcal{Z}}_i, \Omega_i|\boldsymbol{\mathcal{B}})p(\boldsymbol{\mathcal{B}})}{p(\boldsymbol{\mathcal{Z}}_i, \Omega_i)}. \tag{13}$$

However, term $p(\boldsymbol{\mathcal{Z}}_i, \Omega_i|\boldsymbol{\mathcal{B}})$ on the right-hand side (RHS) of Eq. (13) is too complex in nature resulting in complex $p(\boldsymbol{\mathcal{B}}|\boldsymbol{X}_i, \Omega_i)$, which eliminates the possibility of direct evaluation in closed-form. Alternatively, particle-based approximation techniques [24] are utilized to address this issue. In brief, a set of N_p particles $\{\mathbb{B}(k)\}_{k=1}^{N_p}$ generated from the prior density $p(\boldsymbol{\mathcal{B}})$ are utilized, where $\mathbb{B}(k)$ denotes a particle representing a single filter bank. To be more specific, each particle contains the weight of itself ($\pi(k)$) and the characteristics of the spectral filter that it defines ($\mathbb{B}(k) = \{b_s(k), b_e(k), \pi(k)\}$). Hence, we can rewrite the particle-based posterior distribution (Eq. (13)) as

$$p(\mathbb{B}(k)|\boldsymbol{\mathcal{Z}}_i(k), \Omega_i) = \frac{p(\boldsymbol{\mathcal{Z}}_i(k), \Omega_i|\mathbb{B}(k))p(\mathbb{B}(k))}{p(\boldsymbol{\mathcal{Z}}_i(k), \Omega_i)}. \tag{14}$$

The posterior $p(\mathbb{B}(k)|\boldsymbol{\mathcal{Z}}_i(k), \Omega_i)$ provides all the required information regarding $\boldsymbol{\mathcal{B}}$ which can be obtained from the bandpass-filtered signal $\boldsymbol{\mathcal{Z}}_i(k)$ and its corresponding class label Ω_i. The spectral filtering step is then followed by computing the common spatial patterns of each trial in each particle (i.e., $\boldsymbol{\mathcal{Z}}_i(k)$).

Spatial Filtering: This step is performed using the CSP method (Eq. (9)). However, since particles are deployed to solve the optimization problem, we have the spatial filtering coupled with spectral filtering, i.e., Eq. (9) modifies as

$$\boldsymbol{\mathcal{W}}_i^{(j)}(k) = \left[\boldsymbol{W}^{(j)}(k)\right]^T \boldsymbol{\mathcal{Z}}_i(k). \tag{15}$$

Please note that from this stage we add the superscript "$(.)^{(j)}$"$(1 \leq^{(j)} \leq N_{\text{Bits}})$ to the formulas to demonstrate that the ECOC formulations are now in effect. Eq. (15) performs the spatial filtering of the spectrally filtered signals and prepares them for the feature extraction step which is introduced in Eq. (10). After calculating the features for each trial based on the characteristics of each particle, we form a feature matrix which contains all the features for all the trials as follows

$$\boldsymbol{F}^{(j)}(k) = \{\boldsymbol{f}_{i,p}^{(j)}(k)\}_{i=1}^{N_{\text{Trial}}} \in \mathbb{R}^{2m \times N_{\text{Trial}}}. \tag{16}$$

Since the spectra-spatial filtering is a deterministic procedure and does not add any stochastic behavior to the system, we can rewrite the posterior probability in Eq. (14) as follows

$$\begin{aligned} p(\mathbb{B}(k)|\boldsymbol{\mathcal{Z}}_i(k), \Omega_i) &\triangleq p(\mathbb{B}(k)|\boldsymbol{F}^{(j)}(k), \boldsymbol{\Omega}) \\ &= \frac{p(\boldsymbol{F}^{(j)}(k), \boldsymbol{\Omega}|\mathbb{B})p(\mathbb{B}(k))}{p(\boldsymbol{F}^{(j)}(k), \boldsymbol{\Omega})}. \end{aligned} \tag{17}$$

In order to compute the likelihood probability $p(\boldsymbol{F}^{(j)}(k), \boldsymbol{\Omega}|\mathbb{B})$ in Eq. (17), an information-theoretic approach is deployed. In this approach, the mutual information [26] is utilized to compute the likelihood problem and measure the discriminative power of the features in classifying the EEG trials. In this regard, the likelihood is defined as

$$p(\boldsymbol{F}^{(j)}(k), \boldsymbol{\Omega}|\mathbb{B}) = \exp\left[I\big(\boldsymbol{F}^{(j)}(k); \boldsymbol{\Omega}\big)\right], \tag{18}$$

where $I(.)$ denotes the mutual information between the features matrix and the class labels. Please note that the mutual information is actually calculated between the columns of the features matrix (i.e., each feature vector) and the class labels, therefore, it can reveal how much each feature vector is representing the class labels. The mutual information is computed based on the entropy of the features and the labels, i.e.,

$$I(\boldsymbol{F}^{(j)}(k);\boldsymbol{\Omega}) = H(\boldsymbol{F}^{(j)}(k)) - H(\boldsymbol{F}^{(j)}(k)|\boldsymbol{\Omega}). \tag{19}$$

where $H(.)$ and $H(.|.)$ represent the entropy and conditional entropy, respectively [26]. Once the posterior probabilities are estimated, the weights of the particles need to be derived. The weight of each particle is computed as follows

$$\pi^{(j)}(k) = \frac{p(\boldsymbol{F}^{(j)}(k),\boldsymbol{\Omega}|\mathbb{B}_{\mathbf{k}})}{\sum_{k=1}^{N_p} p(\boldsymbol{F}^{(j)}(k),\boldsymbol{\Omega}|\mathbb{B}(\mathbf{k}))}. \tag{20}$$

For each iteration of the optimization procedure, the set of particle weights are calculated according to Eq. (20). Then, the set $\boldsymbol{\Pi}^{(j)}$ is formed according to the following criteria

$$\boldsymbol{\Pi}^{(j)} = \bigcup_{k}(\pi^{(j)}(k) > \tau). \tag{21}$$

The criteria introduced in Eq. (21) selects the particles which have weight ($\pi^{(j)}(k)$) greater than τ, which is a random number between 0 and 1, and in each iteration its value changes. If a particle is selected and is included in the set $\boldsymbol{\Pi}^{(j)}$, its associated characteristics of the filter bank remain intact for the next iteration. However, for those particles which are not included in this set, a disturbance following a normal distribution ($\sim \mathcal{N}(0,1)$) is added to the values of the band limits, which define the characteristics of the filter banks. In the next iteration, the effect of the disturbance (changing the characteristics of the filter banks) is evaluated, and the same procedure goes on for a specific number of iterations until the particles converge to the optimum value of the band limits for the spectral filter and the particle weights. In the final step, after deriving the optimized particles, N_p number of support vector machines (SVMs) classifiers are trained based on the features extracted from each particle.

In the evaluation phase, the unlabeled trial is preprocessed and then processed according to the spectral and spatial filters which are obtained in the training phase. In other words, since we have optimized N_p number of particles, N_p different spectral filters are applied to the trials and the process is followed by applying spatial filters corresponding to each particle. Then the features of each particle are obtained and are fed into the corresponding classifier. To merge the decisions of N_p classifiers, the output of each classifier ($+1$ or -1) is multiplied by the weight of the particle and the final decision of the ECCSP-TB is based on the weighted summation of the outputs of the classifiers. Please note that in order to obtain a smooth classification scheme in the evaluation part, the two classes of data are labeled with $+1$ and -1. As stated above, this labeling comes in handy for merging the decision of the classifiers by incorporating the weight of the particles. Finally, we denote the classifiers' set for all particles as $\{\zeta(k)\}_{k=1}^{N_p}$, therefore, the decision rule in the ECCSP-TB is defined as

TABLE I
GENERATED TERNARY CODEWORDS FOR CLASSIFYING 4 CLASSES OF DATA IN TECOC APPROACH.

Class Label	Assigned class codeword in ternary system
Tongue MI	1 0 1 2 1 0 1 2 0 1 2 0 1 2 1 0 1 2 1 1 0 1 2 1 1
Foot MI	0 1 1 1 2 0 0 0 1 1 1 2 2 2 0 1 1 1 2 0 1 1 1 2 0
Left hand MI	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 2 2 2 2
Right hand MI	0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 0 0 0 0 0 2

$$\lambda_{test}^{(j)} = \text{sign}\big(\sum_{k=1}^{N_p} \pi^{(j)}(k)\zeta(k)(\boldsymbol{X}_{test})\big). \tag{22}$$

This completes feature extraction of the ECCSP-TB variant, next we present the modification of the classification component in comparison to the primary ECCSP framework.

B. Ternary ECOC

As stated previously, the ECOC technique splits the problem of multi-class classification to a number of binary subclassification problems. The action of how to split the problem, requires generating proper codewords. Up until now, deployment of binary codewords is being discussed; however, we propose to extend the ECCSP by generating codewords based on the ternary numerical system.

Intuitively speaking, we believe that the drawback of the ECOC method, introduced in section II, is that in each of the cases that we form the super-sets ($S^{(j,0)}$ and $S^{(j,1)}$), all the classes are involved and there is no chance to examine the classifier $\boldsymbol{\gamma}^{(j)}$ in absence of features of a certain class. Therefore, we propose the Ternary ECOC classifiers, where the possible labels are now 0, 1 and 2 (ternary system). In this work, we take 0 and 1 as labels to form the super-sets, and the classes with label 2, are not involved in the classifier design. More specifically, in the ternary system and in terms of our running example with 4 classes of data (Fig. 2), we can form 81 different numbers ($3^4 = 81$). By performing a similar procedure to that of the conventional ECOC technique for removing the complements and meaningless cases, and taking this fact into account that label "2" means that the class is not involved into the classifier training, we derive 25 particular cases. It is also worth mentioning that any 4-digit ternary number which does not consist of either 0 or 1, is defined as meaningless. Table I demonstrates the ternary codewords which are derived for classification of 4 classes of data based on our running example.

The advantage of the proposed Ternary ECOC over its conventional method is that the Hamming distance between the class codewords is increased, and as a result, the chance of misclassification is reduced. On the other hand, a higher number of trained classifiers and increased computational cost are the drawbacks of this technique. The decision rule for Ternary ECOC is the same as the one in Eq. (2), keeping in mind that the code words here are the ternary ones. Algorithm 3 outlines the training phase of the proposed ECCSP-TB framework, while Algorithm 4 demonstrates the work-flow in the evaluation phase.

IV. THE ECCSP-TB2B

In the previous section, we developed the ECCSP-TB variant, which aims at deriving the optimized spectral filters to bet-

Algorithm 3 THE ECCSP-TB IN TRAINING PHASE

Input: EEG recordings $\{\boldsymbol{X}_i\}_{i=1}^{N_{\text{Trial}}}$ and their corresponding labels $\{\Omega_i\}_{i=1}^{N_{\text{Trial}}}$

Output: N_p number of optimized particles $\{\mathbb{B}(k)\}_{k=1}^{N_p}$ for each bit of (N_{Bits} in total) the ECOC classifier.

for The number of bits in the ECOC codeword (N_{Bits}) **do**
  Construct the two super-sets $\boldsymbol{S}^{(j,0)}$ and $\boldsymbol{S}^{(j,1)}$
  Define N_p number of particles which $\forall k : \pi(k) = \frac{1}{N_p}$.
  Randomly initialize the filter bank band limits.
  for The number of iterations **do**
    for $k \in$ the number of particles **do**
      Spectrally filter the signals by Eq. (11)
      Derive the CSP filters for each frequency band
      Spatially filter the signals by Eq. (9)
      Extract features by Eq. (10)
    end for
    Form the features matrix $\boldsymbol{F}$ by Eq. (16)
    Compute the posterior probability for Eq. (17)
    Calculate the particle weights using Eq. (20)
    if $\pi(k) < \tau$ **then**
      Add normal noise to $[b_s, b_e]$ in particle k.
    end if
  end for
  Train N_p number of SVM classifiers based on the features matrix $\boldsymbol{F}(k)$, from the last iteration.
end for

Algorithm 4 THE ECCSP-TB IN EVALUATION PHASE

Input: Unlabeled EEG Trial $\boldsymbol{X}_{\text{Test}}$, N_p number of trained binary classifiers, optimized particles $\{\mathbb{B}(k)\}_{k=1}^{N_p}$ and spatial filters $\boldsymbol{W}_l(k)$.

Output: The predicted class label for unseen trial $\boldsymbol{X}_{\text{Test}}$.

for The number of particles N_p **do**
  Spectral filtering of the trial based on Eq. (11).
  Spatial filtering of the trials according to Eq. (9).
  Extract classifier-specific features using Eq. (10).
  Obtain the prediction of each classifier for each set of features, λ .
end for

S2. Compute the weighted summation of the predictions.

S3. Obtain the final decision based on the sign of the weighted summation.

ter discriminate the EEG trials for different tasks. In the simulations results, we will show its superior performance among different competing techniques. In order to further boost the performance of the proposed ECCSP framework, in this section, we develop the ECCSP-TB2B variant. Here, the Bayesian framework developed in Section III is incorporated to optimize the characteristics of a filterbank (2 frequency bands), instead of a single spectral filter used in the ECCSP-TB variant. The motivation for this modification is to combine the benefits of the filterbank method and the optimized methods, in a way to reach higher accuracies. The reason for selecting two frequency bands is that we believe that the motory related features of the EEG signals are mainly stored in the μ band ($8-13$Hz) and the β band ($13-30$Hz). Ideally speaking, it is expected that the limits of the incorporated filterbank converge such that the optimized limits maximize the useful extracted frequency contents of the aforementioned bands. In this regard, we initialize a filterbank consisting of 2 bandpass filters, and then we optimize the frequency limits based on the procedure discussed in Section III.

More specifically, in the ECCSP-TB2B variant, the randomness of the cut-off frequencies for the filterbank are modeled with *a priori* probability denoted by $p(\boldsymbol{\mathcal{B}}^{\mathcal{D}})$ over random variable $\boldsymbol{\mathcal{B}}^{\mathcal{D}}$. In this scheme, three random variables are deployed to characterize the filterbank and $\boldsymbol{\mathcal{B}}^{\mathcal{D}}$ is defined as

$$\boldsymbol{\mathcal{B}}^{\mathcal{D}} \triangleq [b_s, b_m, b_e], \tag{23}$$

where ideally the μ band contents are to be extracted with the $[b_s, b_m]$ bandpass filter and the β band contents to be extracted with the $[b_m, b_e]$ bandpass filter. Similar to the ECCSP-TB variant, the posterior density denoted now by $p(\boldsymbol{\mathcal{B}}^{\mathcal{D}}|\boldsymbol{X}_i, \Omega_i)$ is constructed from single-trial EEG recording $\boldsymbol{X}_i$, for ($1 \leq i \leq N_{\text{Trial}}$), and its particle-based approximation is computed as

$$p(\mathbb{B}^{\mathcal{D}}(k)|\boldsymbol{X}_i, \Omega_i) = \frac{p(\boldsymbol{X}_i, \Omega_i|\mathbb{B}^{\mathcal{D}}(k))p(\mathbb{B}^{\mathcal{D}}(k))}{p(\boldsymbol{X}_i, \Omega_i)}, \tag{24}$$

where $\mathbb{B}^{\mathcal{D}}(k) = \{b_s(k), b_m(k), b_e(k), \pi(k)\}$ denotes particle k, for ($1 \leq k \leq N_p$), and represents a single filterbank.

The bandpass filtering of the signals is modeled as a convolution operation on the input signals performed by the system $h_l(k)$, for ($l \in \{1, 2\}$), associated with each of the two bandpass filters. Please note that $l = 1$ determines the filter for $[b_s, b_m]$, and $l = 2$ denotes the filter for $[b_m, b_e]$. Therefore, the filtered signal, denoted by $\boldsymbol{\mathcal{Z}}_l$, is deterministically obtained as

$$\boldsymbol{\mathcal{Z}}_{i,l}(k) = h_l(k) \circledast \boldsymbol{X}_i, \tag{25}$$

The likelihood and the evidence are, therefore, become equal to $p(\boldsymbol{\mathcal{Z}}_{i,l}(k), \Omega_i|\mathbb{B}^{\mathcal{D}}(k))$ and $p(\boldsymbol{\mathcal{Z}}_{i,l}(k), \Omega_i)$, respectively. Hence, we can rewrite (24) by replacing the raw EEG signal $\boldsymbol{X}_i$ with its bandpass-filtered version $\boldsymbol{\mathcal{Z}}_{i,l}(k)$ as

$$p(\mathbb{B}^{\mathcal{D}}(k)|\boldsymbol{\mathcal{Z}}_{i,l}(k), \Omega_i) = \frac{p(\boldsymbol{\mathcal{Z}}_{i,l}(k), \Omega_i|\mathbb{B}^{\mathcal{D}}(k))p(\mathbb{B}^{\mathcal{D}}(k))}{p(\boldsymbol{\mathcal{Z}}_{i,l}(k), \Omega_i)}. \tag{26}$$

The spectral filtering step is then followed by computing the common spatial patterns of each trial for each frequency band in each particle (i.e., $\boldsymbol{\mathcal{Z}}_{i,l}(k)$). The spatial filtering formula is rewritten as $\boldsymbol{\mathcal{W}}_{i,l}^{(j)}(k) = \left[\boldsymbol{W}_l^{(j)}(k)\right]^T \boldsymbol{Z}_{i,l}(k)$, and the feature extraction formula is updated as

$$\boldsymbol{f}_{i,l}^{(j)}(k) = \log\left(\frac{\text{var}\big(\boldsymbol{\mathcal{W}}_{i,l,p}^{(j)}(k)\big)}{\sum_i \text{var}\big(\boldsymbol{\mathcal{W}}_{i,l,p}^{(j)}(k)\big)}\right). \tag{27}$$

Now, at this stage, the features of the two parallel pipelines are merged into one single feature vector as

$$\boldsymbol{f}_i^{(j)}(k) = \left[f_{i,l}^{(j)}(k)|_{l=1}, \boldsymbol{f}_{i,l}^{(j)}(k)|_{l=2}\right], \tag{28}$$

and then the vectors are combined according to Eq. (16). The rest of the procedure is the same as discussed in Section III.

V. Simulations

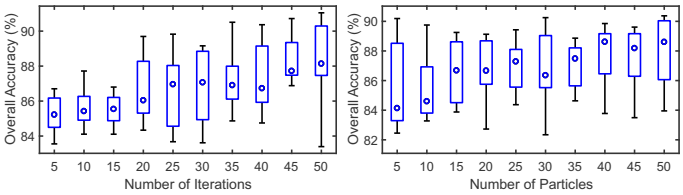
In this section, we present an extensive set of simulation experiments to evaluate the performance of the proposed ECCSP framework and its variants. The experiments are performed based on two well regarded MI datasets provided by BCI Competition IV, referred to as "Dataset BCIC-IV_{2a}" and "Dataset BCIC-IV_{2b}" [25]. The former dataset consists of EEG signals corresponding to the multi-class classification problem of 4 MI tasks and is selected to quantify the performance of the proposed ECCSP framework, i.e., when the (Ternary) ECOC classifiers are coupled with the optimized spatio-spectral filter-banks. The latter dataset considers the problem of binary task classification and is selected to evaluate the performance of the feature extraction module of the proposed ECCSP framework.

Dataset BCIC-IV_{2a}: This dataset consists of four MI classes of EEG measurements (Right hand MI, Left hand MI, Foot MI, and Tongue MI), obtained from 9 healthy subjects. Signals are recorded with sampling rate of 250Hz, using 22 EEG channels and 3 monopolar electrooculogram (EOG) channels (with left mastoid serving as the reference). The original EEG signal recordings are already bandpass filtered ($0.5 - 100$Hz) and notch filtered to remove the interference of the power line on the signals. For each subject, two sessions are recorded (one for training purposes and the other one for evaluation). Each session consists of six segments, and each segment includes 48 trials of length 3 seconds. In total and for each subject, 288 trials for training and 288 trials for evaluation are available. In order to measure the performance of the proposed ECCSP framework and based on the recommendation from BCI competition [25], kappa coefficient κ is used, i.e., $\kappa = \frac{\text{CCR} - P_{\text{rand}}}{1 - P_{\text{rand}}}$, where CCR represents the correct classification rate, and the value of P_{rand} for this dataset is equal to 0.25. All the algorithms are implemented based on the 22 channel recordings and the EOG channels are not used.

Dataset BCIC-IV_{2b}: This dataset consists of the EEG recordings from 9 different right-handed subjects, with normal or corrected-to-normal vision. The subjects were asked to perform two MI tasks including "Left Hand MI" and "Right Hand MI." From each subject, 5 sessions are recorded, 3 of them for training and the rest for evaluation. Among the training sessions, 2 of them are recorded without providing feedback to the subject, and one is recorded when the feedback is enabled. This dataset provides 6 recording channels, 3 for EEG and 3 for EOG. The EEG channels are recorded form C3, Cz, and C4 points in the 10-20 EEG recording system, similar to the previous dataset we only use EEG recordings. The signals are recorded with sampling frequency of 250Hz and bandpass filtered between $0.5 - 100$Hz. A notch filter is also applied to remove the 50Hz effect of the power line on the recordings. For this dataset, the performance measure is also kappa coefficient where the probability of arbitrary classification is 50%.

A. Parameter Selection

Similar in nature to particle filter based approaches [24], we need to select the number of particles and the number of optimization iterations to be used during the training phase of the proposed ECCSP-TB and ECCSP-TB2B variants. In this regard and to utilize reasonable values for these parameters, we have conducted two experiments computing the 10×10-fold cross-validation as a measure of performance with respect to the changes in the aforementioned parameters. It is expected that these two parameters jointly affect the performance of the system and the problem of finding the optimum values should be solved while we simultaneously change the parameters together, however, addressing the 2 parameter optimization problem in this fashion imposes extensive computational costs. Consequently, our experiments are based on a naive hypothesis which tests the effects of these parameters on the overall performance independent of each other. This results in a simplified optimization problem which performs grid search on a certain range of values for the parameters. We would like also to note that in the optimization stage, a stochastic process is in effect as the particles are being updated based on a Gaussian distribution, therefore, we expect the performance of the system to change slightly, even with the fixed parameters. To eliminate the effect of this randomness in our parameter selection process, we have performed each experiment for 10 times and the mean and variance of the performance is presented in box plots. Parameter selection step is performed based on the EEG recordings from subject 1 in dataset BCIC-IV_{2b} as detailed below:



(a) (b)

Fig. 4. 10×10-fold cross-validation. (a) Different number of iterations: 45 iterations are selected as the optimum one. (b) Different number of particles: 40 particles are selected as the optimum one.

Number of Iterations: In this experiment we investigate the number of iterations which yield the maximum cross-validation performance. To this aim, we keep the number of particles constant equal to 30 particles. This number is reported by [21], which was the first paper proposing the utilization of a Bayesian framework for spatial-spectral filter optimization. Then the number of iterations is varied from 5 to 50 with steps of 5, and the 10×10-fold cross-validation of the performance is measured. The results are presented in Fig. 4(a). It is observed that the number of iterations providing the best performance in terms of maximum mean and minimum variance is 45. We note that, although for 50 number of iterations higher mean performance is achieved, the variance the range of results is wider making it undesirable.

Number of Particles: Here, the number of iterations is set to 45 and we vary the number of particles within the range of 5 to 50 with steps of 5 to select the best option for the number of particles. Likewise the previous experiment, 10×10-fold cross-validation is deployed to measure the performance of the system. The results of this experiment are presented in Fig. 4, which suggests that using 40 particles results in the maximum mean performance and minimum variability across different runs. In all of the following experiments, we selected 40 number of particles and 45 number of optimization iterations

for evaluation of the ECCSP-TB and ECCSP-TB2B variants.

B. Experimental Procedure, Results, and Discussions

Segments of 2 seconds (500 samples) from each trial are selected as the input to each framework. The segment starts 0.5 seconds after the onset cue of each trial and lasts for 2 seconds. For the spectral filters which are used in all of the algorithms, we have deployed 5th order Butterworth bandpass filter. Since the optimization procedure is time-consuming and computationally extensive, we decided to deploy a low order spectral filter but kept it the same across all the algorithms for fair comparison. For the spatial filtering of the signals, we have utilized the CSP technique in its most basic form. The reason for this selection is that our main objective is to demonstrate the effectiveness of deploying optimized spectral filterbanks on the overall performance of the system, therefore, we used basic forms of the spatial filtering and feature extraction.

After spatial filtering of the signals, the first and last m rows of the signals are used for feature extraction. For dataset IV_{2a} we have $m = 2$ and for dataset IV_{2b} we have $m = 1$. In the classification step, the SVM is used as the underlying binary classifier of the ECOC classifiers. It is worth mentioning that the SVM classifier in its simplest format is deployed. In the training phase and according to the procedure described in Section II, we deploy generated codewords for the 4 class classification problem which are introduced in Fig. 3 and Table I. When it comes to the evaluation phase, the extracted test trials are fed into each trained particle and are filtered according to the specific spectral characteristics of that particle. The votes of the particles are then integrated into the final decision based on Eq. (22). This whole process produces one bit out of the N_{Bits} required for the Ternary ECOC classifier to produce the final decision based on Eq. (2).

To evaluate the performance of the proposed ECCSP framework and its variants, we compare the results with successful and state-of-the-art MI algorithms. Results from eight different approaches are computed and compared for accuracy in Table II. In particular, we have implemented and compared our framework against the results of the following algorithms: (i) The FBCSP method [16], which is the winner of the BCI competition IV and achieved the highest performance for this dataset; (ii) The SCSSP method [19], which is one of the most recent works on this problem, and; (iii)-(iv) The BSSFO [21] framework when coupled with the proposed Ternary ECOC and conventional ECOC classifiers. In addition, we have provided the simulation results for the two techniques which are served as extensions of binary classifiers for multi-class classification problems. The two techniques are called "One-Versus-One (OVO)" and "One-Versus-All (OVA)", which are well extensively used for the multi-class classification problems. The two methods are believed to yield compromising results in extension of binary classifiers to multi-class classifiers. In the OVO approach, for any possible pair of classes, a classifier is trained, and then for an unknown trial, the final decision is made based on the most frequent class label in the results of the classifiers. In a generalized example, for N_c number of classes, $\binom{N_c}{2} = \frac{N_c!}{(N_c-2)!2!}$ number of classifiers should be trained. In the OVA, each class out of N_c classes should be discriminated from a super-set of all other classes, therefore, N_c number of classifiers needs to be trained to form the extension of the binary classifiers to the multi-class classifiers. In our current problem of classifying 4 different classes, we have to train 4 classifiers for the following sets: $[1, \{2, 3, 4\}], [2, \{1, 3, 4\}], [3, \{1, 2, 4\}], [4, \{1, 2, 3\}]$. The final decision could be based on two criteria: (a) The class label which is reported more frequently than others be selected as the assigned label to the unknown trial, or; (b) The label which is reported with higher score be assigned to the unknown trial.

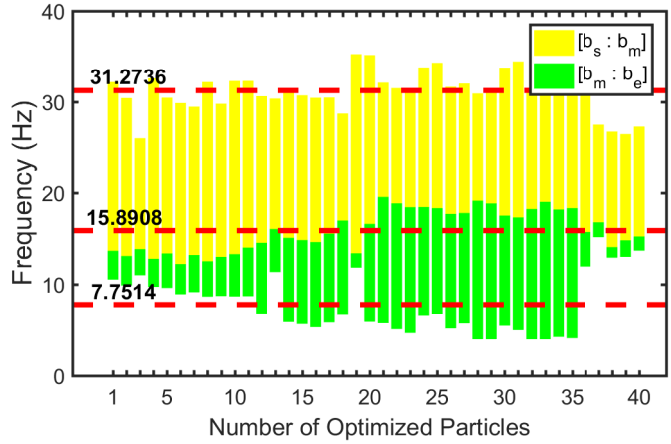


Fig. 5. Optimized frequency bands for all of the particles within 45 iterations.

The performance of the proposed frameworks evaluated on dataset BCIC-IV_{2a} is provided in Table II and the results for dataset BCIC-IV_{2b} is presented in Table III. For the ECCSP-TB and ECCSP-TB2B variants, it is worth mentioning that in the first iteration, the particles are weighted equally $(1/N_p)$, therefore, the initial value for the particle weights is set to $1/40$ and the band limits of the spectral filters are initialized with random numbers within the range of 4-40 Hz. Please note that for all of the techniques, the spectral filter which is used is a 5th order Butterworth bandpass filter. As it is shown in Table II and Table III, the proposed ECCSP-TB2B framework outperforms its variations as well as other existing frameworks. Please note that to maintain the fidelity of the comparison, we have not performed the feature selection procedure which is discussed in the original FBCSP method, and we also have deployed the same SVM classifier for all of the methods. Moreover, the spectral filter that we deployed in this study for the ECCSP and its variations, FBCSP and BSSFO is a $5^{th}-$order Butterworth IIR filter; which is not necessarily the same as the spectral filter which is deployed in the original papers of FBCSP, SCSSP and BSSFO.

As a final experiment, we investigate the values of the frequency bands when the optimization process is finished. As stated previously, our intuition for deploying two frequency bands within the ECCSP-TB2B framework is to extract frequency specific features which are related to μ and β bands. In other words, we hope that the frequency bands in the particles converge to the typical limits of the two aforementioned bands which are $8 - 13$Hz and $13 - 30$Hz, respectively. The results for subject 1 in dataset IV_{2b} is depicted in Fig 5. As it is shown in the figure, the means of the parameters b_s, b_m and

TABLE II
PERFORMANCE COMPARISON FOR DIFFERENT APPROACHES, TESTED ON $BCIC - IV_{2a}$ DATASET. PERFORMANCE MEASURE IS IN KAPPA (κ) VALUE.

Subjects	**ECCSP**	**SCSSP** [19]	**FBCSP** [19]	**BSSFO OVO**	**BSSFO OVA**	**BSSFO Conventional ECOC**	**ECCSP-TB**	**ECCSP-TB2B**
Subject 1	48.14	62.25	68.07	62.04	31.02	51.85	59.26	**75.46**
Subject 2	20.37	26.54	30.25	19.91	7.87	14.35	24.54	**32.41**
Subject 3	49.07	65.76	70.68	70.37	56.94	56.48	64.35	**79.63**
Subject 4	31.49	26.91	39.34	40.74	45.37	53.7	50.46	**53.7**
Subject 5	16.2	6.64	**28.42**	9.26	-7.4	6.94	15.28	18.06
Subject 6	12.5	25.56	25.01	14.81	8.33	17.59	23.61	**30.55**
Subject 7	63.88	40.65	56.89	71.76	62.04	63.89	74.54	**78.24**
Subject 8	38.42	59.32	59.13	57.87	30.09	52.31	63.89	**69.44**
Subject 9	37.96	65.86	54.06	65.28	54.63	56.48	68.52	**74.54**
Average	35.34	42.17	47.98	45.78	32.1	41.51	49.38	**56.89**

TABLE III
PERFORMANCE COMPARISON FOR DIFFERENT APPROACHES, TESTED ON BCIC-IV_{2b} DATASET. PERFORMANCE MEASURE IS IN KAPPA (κ) VALUE.

Subjects	**ECCSP**	**ECCSP-TB**	**FBCSP**	**ECCSP-TB2B**
Subject 1	15	19.44±5.64	21.25	28.13±5.46
Subject 2	1.43	10.36±2.19	15.71	15±3.87
Subject 3	23.75	7.31±2.8	-4.38	8.75±5.13
Subject 4	36.88	91.69±2.21	61.25	95.63±2.11
Subject 5	23.75	19.56±5.15	55	57.5±6.43
Subject 6	-5.63	44.94±10.6	23.13	60.63±4.7
Subject 7	44.38	44.63±3.44	30.63	55.63±5.25
Subject 8	74.38	77.31±0.84	17.5	79.38±3.19
Subject 9	30.63	61.94±2.56	13.13	64.38±3.32
Average	27.17	41.91	25.91	51.67

b_e are rather close to the typical limits of μ and β bands, which approves our intuition that we need to deploy two frequency bands for the filterbank, and that the optimization framework works as expected.

VI. CONCLUSION

Extracting discriminative features from the EEG signals plays an important role in the performance of any EEG-based BCI system. Different factors contribute to properness of the features including the preprocessing step, the spectral filters, the spatial filters, and the feature extraction technique. In this work, we demonstrated the effect of the spectral filters on the overall performance of the system. To this aim, we deployed a Bayesian framework to optimize the characteristics of the spectral filters, in order to best fit them for each subject's motory behavior. In addition, we proposed the integration of the ECOC classifier and its modified version, called *Ternary*-ECOC, with the introduced optimization framework to fit the framework for multi-task MI classification problems.

REFERENCES

[1] S. Makeig, *at al.*, "Evolving Signal Processing for BrainComputer Interfaces," *Proc. IEEE*, vol. 100, pp. 1567-1584, 2012.

[2] K.R. Muller, B. Blankertz, "Toward Noninvasive Brain-Computer Interfaces," *IEEE Signal Process. Mag.*, vol. 23, no. 5, pp. 128-126, 2006.

[3] A.S. Aghaei, *at al.*, "Smart Driver Monitoring: When Signal Processing Meets Human Factors: In the Driver's Seat," *IEEE Signal Process. Mag.*, vol. 33, no. 6, pp. 35-48, 2016.

[4] G. Schirner, D. Erdogmus, K. Chowdhury, T. Padir, "The Future of Human-in-the-Loop Cyber-Physical Systems," *Computer*, vol. 46, no. 1, pp. 36-45, 2013.

[5] J.J Daly, J.R. Wolpaw, "Brain-Computer Interfaces in Neurological Rehabilitation," *The Lancet Neurology*, vol. 7, no. 11, pp. 1032-1043, 2008.

[6] Ren Xu, *et al.*, "A Closed-Loop Brain-Computer Interface Triggering an Active Ankle-Foot Orthosis for Inducing Cortical Neural Plasticity," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 7, pp. 2092-2101, 2014.

[7] M. Grosse-Wentrup, D. Mattia, and K. Oweiss, "Using Brain-Computer Interfaces to Induce Neural Plasticity and Restore Function," *Journal of Neural Engineering*, vol. 8, no. 2, 2011.

[8] R. Leeb, L. Tonin, M. Rohm, L. Desideri, T. Carlson, J.D.R. Millán, "Towards Independence: A BCI Telepresence Robot for People with Severe Motor Disabilities," *Proc. IEEE*, vol. 103, no. 6, pp. 969-982, 2015.

[9] L.V. Dokkum, T. Ward, I. Laffont, "Brain Computer Interfaces for Neurorehabilitation: Its Current Status as a Rehabilitation Strategy," *Annals Physical & Rehabil. Medicine*, vol. 58, no. 1, pp. 3-8, 2015.

[10] O. Talakoub, C. Marquez-Chin, M.R. Popovic, J. Navarro, E.T. Fonoff, C., Hamani, and W. Wong, "Reconstruction of Reaching Movement Trajectories using Electrocorticographic Signals in Humans," *PLoS ONE*, vol. 12, no. 9, 2017.

[11] B.J. Edelman, B. Baxter, and B. He, "EEG Source Imaging Enhances the Decoding of Complex Right-hand Motor Imagery Tasks," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 1, pp. 4-14, 2016.

[12] L. Yao, X. Sheng, D. Zhang, N. Jiang, N. Mrachacz-Kersting, X. Zhu, and D.Farina, "A Stimulus-Independent Hybrid BCI based on Motor Imagery and Somatosensory Attentional Orientation," *IEEE Trans. Neural Syst. Rehabil. Eng.*, 2017.

[13] H. Ramoser, J. Muller-Gerking, and G. Pfurtscheller, "Optimal Spatial Filtering of Single Trial EEG During Imagined Hand Movement," *IEEE Trans. Rehabil. Eng.*, vol. 8, no. 4, pp. 441-446, 2000.

[14] X. Xiaofeng, *et al.* "Motor Imagery Classification based on Bilinear Sub-manifold Learning of Symmetric Positive-Definite Matrices," *IEEE Trans. Neural Syst. Rehabil. Eng.*, 2016.

[15] S. Shahtalebi, A. Mohammadi: "Error Correction Output Codding Coupled with the CSP for Motor Imagery BCI Systems," *European Signal Processing Conference*, 2017, pp. 2071-2075.

[16] K.K. Ang, Z.Y. Chin, C. Wang, C. Guan, and H. Zhang, "Filter Bank Common Spatial Pattern Algorithm on BCI Competition IV Datasets 2a and 2b," *Frontiers in Neuroscience*, vol. 6, 2012.

[17] H. Lu, H.L. Eng, C. Guan, K.N. Plataniotis, and A.N. Venetsanopoulos, "Regularized Common Spatial Pattern with Aggregation for EEG Classification in Small-sample Setting," *IEEE Trans. Biomed. Eng.*, vol. 57, no. 12, pp. 2936-2946, 2010.

[18] Fabien Lotte and Cuntai Guan, "Regularizing Common Spatial Patterns to Improve BCI Designs: Unified Theory and New Algorithms," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 2, pp. 355–362, 2011.

[19] A.S. Aghaei, M.S. Mahanta, and K.N. Plataniotis, "Separable Common Spatio-spectral Patterns for Motor Imagery BCI Systems," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 1, pp. 15-29, 2016.

[20] X. Li, C. Guan, H. Zhang, and K.K. Ang, "A Unified Fisher's Ratio Learning Method for Spatial Filter Optimization," *IEEE Trans. Neural Syst. Rehabil. Eng.*, 2016.

[21] H.I. Suk and S.W. Lee, "A Novel Bayesian Framework for Discriminative Feature Extraction in Brain-Computer Interfaces," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 2, pp. 286-299, 2013.

[22] M. Liu, D. Zhang, S. Chen, and H. Xue, "Joint Binary Classifier Learning for ECOC-Based Multi-Class Classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 11, pp. 2335-2341, 2016.

[23] N. Eghbali and G.A. Montazer, "Improving Multiclass Classification using Neighborhood Search in Error Correcting Output Codes," *Pattern Recognition Lett.*, vol. 100, pp. 74-82, 2017.

[24] A. Mohammadi, A. Asif, "Distributed Consensus + Innovation Particle Filtering for Bearing/Range Tracking with Communication Constraints," *IEEE Trans. Signal Process.*, vol. 63, no. 3, pp. 620-635, 2015.

[25] M. Tangermann, *et al.*, "Review of the BCI Competition IV," *Frontiers in Neuroscience*, vol. 6, pp. 55, 2012.

[26] N. Kwak and C. Choi, "Input Feature Selection by Mutual Information based on Parzen Window," *IEEE Trans. Pattern Anal. Mach. Intell.*,

vol. 24, no. 12, pp.1667-1671, 2002.